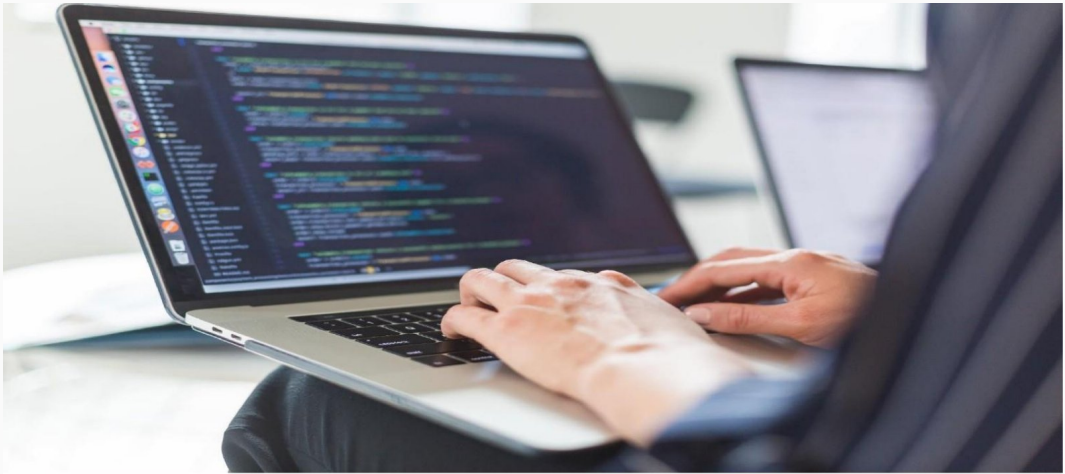


ETX1100/5900 Business Statistics

The Chi-Square Test for Independence: Assessing relationships between categorical variables

The Chi-Square Test for Independence: Assessing relationships between categorical variables

Textbook references:

Berenson et al *Basic Business Statistics 5th edition,*
Chapter 15, Section 15.3.

Assignment

- Due date: 19 January 2025, 11:55 pm
- Weighting: This assignment is worth 15% of the total assessment for this unit.
- Submission location: Submit via the Assignment submission link available under the Assessment tab on Moodle.
- Number of attempts: Only ONE (1) attempt is permitted. •
- Dataset: Use the Excel file titled “Assignment Data.xlsx”, available under the Assessment tab on Moodle.

Hypothesis Test for Evidence-based Decisions

A statistical framework for using data to derive evidence-based decisions.

- Define business problem and variables relevant to that problem
- Formulate a **hypothesis** around these variables that are relevant to business decisions
- Collect representative data
- Conduct hypothesis testing to **establish the degree of evidence** for the hypotheses
- Based on evidence, make business decisions

Hypothesis Testing ……..Is a familiar process

An hypothesis test is conducted to decide between two hypotheses: the null H_0 and the alternative, H_1.

H_0: Defendant is not guilty ('status quo'; default)

H_1: Defendant is guilty (experimental/research hypothesis)



Reject H_0:

There ***IS*** enough evidence against H_0

GUILTY!

Do not reject H_0:

There ***IS NOT*** enough evidence against H_0

NOT GUILTY (INNOCENT!)

Chi Square Test For Independence

We conduct a hypothesis test when we consider relationships between categorical variables - Chi Square Test For Independence

For a contingency table that has r rows and c columns, you can generalise the χ^2 test as a test of independence for two categorical variables.

slido



What types of variables are "Education Level" & "Smoking Status", respectively.

Multiple Choice Poll | 124 votes | 124 participants

Categorical Ordinal, Categorical Nominal - 110 votes ✓

89%

Categorical Nominal, Numerical Discrete - 4 votes

3%

Categorical Nominal, Categorical Ordinal - 10 votes

8%

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If two variables are independent

Multiple Choice Poll | 86 votes | 86 participants

they are not related - 86 votes

100%

they are related - 0 votes

0%

slido

5 Steps

Step 1:

H_0: The two categorical variables are independent (No relationship between variables)
H_1: The two categorical variables are dependent (i.e. there is a relationship).

Note: We test on the assumption that the Null hypothesis (H_0) is true i.e. that the attributes are independent.

What we actually observe in the data
vs
What we would expect to see *if the two variables were independent*

If the observed counts are **very different** from the expected counts, that's evidence

5 Steps

Step 2: Test Statistic

The test statistic is calculated as: $\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$

Where f_o = **observed** frequencies (or count, obtained from the contingency table)

f_e = **expected** frequencies assuming H_0 is true (f_e $\frac{\text{Row Total x Column Total}}{\text{Overall Total}}$)

Notes to remember:

1. *This test statistic has an approximate chi-square distribution with df= (r-1)(c-1)*
2. *The results are not reliable if the expected frequencies are less than 5.*
3. *In this statistical test, we determine whether observed frequencies are significantly different from expected frequencies. If there is a significant difference, we reject the null hypothesis and conclude that the sample data provides sufficient evidence to support the alternative hypothesis.*

5 Steps

Step 3: Critical Value or P-value

Critical value approach/ p-value approach

Critical value approach:

Look up χ^2 table

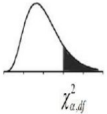
OR

P-value:

Use Excel

=chisq.test(actual range,expected range)

Critical values of the chi square distribution



α:	0.995	0.99	0.975	0.95	0.90	0.10	0.05	0.025	0.01	0.005
df: 1			0.001	0.004	0.016	2.706	3.841	5.024	6.635	7.879
2	0.010	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210	10.597
3	0.072	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345	12.838
4	0.207	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277	14.860
5	0.412	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086	16.750
6	0.676	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812	18.548
7	0.989	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475	20.278
8	1.344	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090	21.955
9	1.735	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666	23.589
10	2.156	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209	25.188
11	2.603	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725	26.757
12	3.074	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217	28.300
13	3.565	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688	29.819
14	4.075	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141	31.319

Steps 4 and 5: Decision and Conclusion

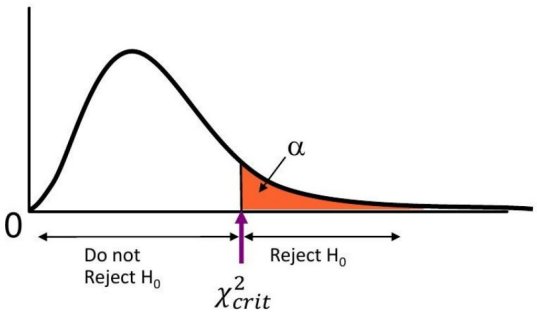
Step 4: Decision

Reject Ho when

$\chi^2_{calc} > \chi^2_{crit}$

OR

Reject Ho when p-value < α



Step 5: Conclusion

State the conclusion within the context of the problem. Always refer to H_1

Eg: *The sample provides sufficient evidence at the 5% level of significance that the*

Task 1: Education Level vs Smoking Status

During the past decade many smokers have attempted to quit. Unfortunately, nicotine is highly addictive. A researcher from the Addiction Research Council wanted to determine why some people quit for good while others relapse. He surveyed 115 people who planned to quit smoking and, among other things, recorded their educational level and whether, one year later, they continued to smoke. The following pivot table was produced, which shows the counts for the various combinations of categories:

Education Level	Smoking Status	
	Did not quit	Quit
Did not finish high school	9	16
High school graduate	26	21
TAFE graduate	4	15

Task 1: Education Level vs Smoking Status

a) What is the proportion of TAFE graduates?

b) We wish to test the hypothesis that Education Level and Smoking Status are independent at a 5% level of significance. What type of test is appropriate? Explain.

c) State the hypotheses for the test stated above.

d) State the test statistic and the p-value.

e) Interpret the results in the context of the question.

Task 1: Education Level vs Smoking Status

a) What is the proportion of TAFE graduates?

Proportion of TAFE Graduates = 19/115 = 0.1652 or 16.52%

b) We wish to test the hypothesis that Education Level and Smoking Status are independent at a 5% level of significance. What type of test is appropriate? Explain.

The Chi-square test of independence is an appropriate test since the variables "Smoking status" and "Education Level" are both categorical variables.

c) State the hypotheses for the test stated above.

H_0: The variables Smoking Status and Education Level are independent

H_1: The variables Smoking Status and Education Level are dependent

Task 1: Education Level vs Smoking Status

d) State the test statistic and the p-value

e) Interpret the results in the context of the question.

Task 1: Education Level vs Smoking Status

d) State the test statistic and the p-value

Test statistic = 8.778 (refer to Excel sheet for working or next slide for screenshot)
P-value = 0.032391

	A	B	C	D	E	F	G
1	Education Level	Smoking Status					
2		Did not quit	Quit			Expected Frequencies	
3	Did not finish high school	9	16	25		10	15
4	High school graduate	26	21	47		18.8	28.2
5	TAFE graduate	4	15	19		7.6	11.4
6	University bachelors or postgraduate degree	7	17	24		9.6	14.4
7		46	69	115			
8							
9	P-value	0.032391					
10	Test Statistic	8.778128					
11	Critical Value	7.814728					
12							

Task 1: Excel formulas

	A	B	C	D	E	F	G
1	Education Level	Smoking Status					
2		Did not quit	Quit			Expected Frequencies	
3	Did not finish high school	9	16	=SUM(B3:C3)		=($D3*B$7/$D$7)	=($D3*C$7/$D$7)
4	High school graduate	26	21	=SUM(B4:C4)		=($D4*B$7/$D$7)	=($D4*C$7/$D$7)
5	TAFE graduate	4	15	=SUM(B5:C5)		=($D5*B$7/$D$7)	=($D5*C$7/$D$7)
6	University bachelors or postgraduate degree	7	17	=SUM(B6:C6)		=($D6*B$7/$D$7)	=($D6*C$7/$D$7)
7		=SUM(B3:B6)	=SUM(C3:C6)	=SUM(B7:C7)			
8							
9	P-value	=CHISQ.TEST(B3:C6,F3:G6)					
10	Test Statistic	=CHISQ.INV.RT(B9,3)					
11	Critical Value	=CHISQ.INV.RT(0.05,3)					
12							

Task 1: Education Level vs Smoking Status

e) Interpret the results in the context of the question.

Since the p-value (0.0324) is smaller than alpha (0.05), we **reject** the null hypothesis.

The sample provides **sufficient** evidence at the **5%** level of significance that the variables Smoking Status and Education Level are **dependent**.

Task 2: Convertibles

It is generally agreed that convertibles offer freedom and excitement. They create a sensation of being a part of your surroundings that no sunroof can offer. On the other hand, sedans are stylish yet practical for small families. The back doors are also handy for rear seat access, and the boot is a secure way to store luggage or cargo.

A car importer believes that there is a relationship between buyers' **gender** and their **preference of sedans or convertibles**. The importer gathered some information on the preference of male and female buyers and this is shown in below.

Task 2: Convertibles

TABLE 1		Gender		
		Male	Female	Total
Preference	Convertibles	30	8	38
	Sedans	56	38	94
Total		86	46	132

We created another contingency table, halved all observed frequencies, and labeled it as Table 2.

TABLE 2		Gender		
		Male	Female	Total
Preference	Convertibles	15	4	19
	Sedans	28	19	47
Total		43	23	66

Task 2: Convertibles

Solutions for part a:

Test statistic for Table 1: **4.4732** p-value for Table 1: **0.0344**

Test Statistic for Table 2: **2.2366** p-value for Table 2: **0.1348**

Task 2: Convertibles

b. Point out why the formula for the χ^2 value allows the test statistic and p-value to be different in the two cases even though all the observed and expected frequencies are proportional.

The formula for the test statistic χ^2 has a square of frequencies in the numerator and just a frequency in the denominator.
So, if you double all the frequencies, you will double χ^2.

Building on this idea:
Doubling all observed and expected frequencies results in the numerator increasing by a factor of four (because of the square) and the denominator increasing by a factor of two.
As a result, the overall χ^2 value doubles. This explains why the test statistic is different, even though the tables are proportional.

Task 2: Convertibles

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Task 3: The Missing Data Mystery



Task 3: The Missing Data Mystery

- There are 4 statistical puzzles!
- Work in your group to **crack each statistical puzzle level by level**.
- Solve the statistical tasks correctly to unlock the next stage.
- After each level, solve a **short riddle** to collect a **keyword. There are also some emojis to help you** answer the riddles.
- Once all levels are completed, **combine the collected words into a final answer**.
- **Accuracy matters more than speed**.
- Good luck, detectives!

Task 3: The Missing Data Mystery

- P(UsedCoupon = Yes) = 0.5
- P(UsedCoupon = Yes | AgeGroup = Young) = 0.67

- Using these probabilities, determine whether the two categories (UsedCoupon and AgeGroup) are dependent or not. Select the correct answer.

A. *Dependent*
B. *Independent*

Task 3: The Missing Data Mystery

- Find the p-value = 0.72

- The evidence has been laid out, and it's time to make a decision. Is there enough evidence to send the Disruptor to jail?

A. We reject the null at the 5% level of significance. The categories are dependent. We can send the Distorter to jail.

B. We do not reject the null at the 5% level of significance. The categories are not dependent. The Distorter goes free.